Implementation and Theoretical Analysis of an End-to-End 5G NR Testbed Using srsRAN and Open5GS

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1. Introduction

The fifth generation (5G) New Radio (NR) system, standardized under 3GPP Release-15, introduces a flexible and scalable air interface along with a cloud-native, service-based core network architecture. The primary objective of 5G NR is to support heterogeneous service requirements including enhanced Mobile Broadband (eMBB), Ultra-Reliable Low Latency Communications (URLLC), and massive Machine Type Communications (mMTC).

From a system perspective, 5G departs from monolithic network design by introducing functional split in the Radio Access Network (RAN) and a modular Service-Based Architecture (SBA) in the 5G Core (5GC). While the standards describe these mechanisms formally, practical implementation is essential to understand protocol interactions, timing dependencies, and cross-layer effects. This project focuses on implementing a complete end-to-end 5G NR testbed using open-source platforms to bridge the gap between specification-level theory and real system behaviour.

2. Overall System Architecture

The implemented system follows the high-level 3GPP Release-15 architecture and consists of the following components:

1) A 5G NR gNB providing radio access and protocol termination
2) A 5G Core Network implementing control-plane and user-plane functions
3) A User Equipment (UE) initiating registration and data sessions

The gNB and core network communicate using standardized NG interfaces, while the UE interacts with the network through RRC and NAS signaling. This separation of concerns allows independent evolution and scaling of RAN and core components.

3. Radio Access Network Implementation

A. *gNB Configuration using srsRAN*

The gNB is implemented using srsRAN, which provides a reference implementation of the NR protocol stack. The gNB terminates the PHY, MAC, RLC, PDCP, and RRC layers as defined in 3GPP specifications. Key theoretical concepts implemented include slot-based scheduling, HARQ processes, logical channel multiplexing, and RRC state transitions.

B. *ZMQ-based RF Interface*

To decouple protocol validation from radio channel effects, a ZMQ-based RF interface is employed. This interface operates at the baseband sample level and enables deterministic testing of higher-layer procedures such as RRC connection establishment and NAS message exchange. This setup is particularly useful for theoretical validation of protocol sequences without stochastic RF impairments.

C. Over-the-Air RF using SDR

For realistic evaluation, the system is extended to over-the-air operation using Software Defined Radio platforms such as USRP and BladeRF. This introduces practical aspects including noise, frequency offset, timing misalignment, and hardware non-idealities, allowing comparison between theoretical expectations and real-world performance.

4. 5G Core Network Implementation

A. Service-Based Architecture

The 5G Core Network is implemented using Open5GS and follows the Service-Based Architecture paradigm. Core functions are decomposed into network functions communicating over well-defined service interfaces. This design improves scalability, fault isolation, and flexibility compared to legacy EPC architectures.

B. Core Network Functions

The following core network functions are instantiated:

1) Access and Mobility Management Function (AMF) for registration and mobility
2) Session Management Function (SMF) for PDU session control
3) User Plane Function (UPF) for packet forwarding

Each function maintains well-defined state machines as described in the 3GPP specifications, enabling modular control-plane operation.

C. NGAP and NAS Signaling

The NG Application Protocol (NGAP) enables signaling between the gNB and AMF, while Non-Access Stratum (NAS) signaling operates between the UE and the core. Theoretical NAS procedures such as authentication, security mode control, and registration acceptance are verified through log-level analysis.

5. Protocol Stack Analysis

A. Physical Layer Considerations

The PHY layer implements OFDM-based waveform generation, slot timing, and modulation schemes. Observations are made regarding synchronization stability, resource block allocation, and link robustness under varying conditions.

B. MAC and RLC Layer behaviour

The MAC layer scheduler determines resource allocation based on buffer status and channel conditions, while HARQ mechanisms ensure reliability. The RLC layer provides segmentation, reassembly, and retransmission services, forming a bridge between PHY reliability and higher-layer requirements.

C. PDCP and RRC Procedures

PDCP handles sequence numbering and header processing, while RRC governs control plane signaling, configuration management, and state transitions. These layers represent the interface between radio-specific behaviour and core-network procedures.

6. Session Establishment and User Plane

A. PDU Session Establishment

Following successful registration, PDU session establishment is initiated. This process involves coordination between AMF and SMF, allocation of user-plane resources, and creation of GTP-U tunnels between the gNB and UPF.

B. User Plane Data Flow

User-plane traffic flows through the gNB and UPF using GTP-U encapsulation. Data throughput, packet delivery, and retransmission behaviour are analyzed to study the impact of lower-layer scheduling and channel conditions.

7. Debugging and Validation

A. Failure Case Analysis

Several failure scenarios are analyzed to understand system robustness:

1) Registration failures due to incorrect configuration or timing mismatch
2) PDU session failures caused by control-plane inconsistencies
3) Throughput degradation linked to scheduler behaviour

B. Cross-Layer Log Correlation

Logs from PHY, MAC, RLC, PDCP, RRC, NGAP, and NAS layers are correlated to trace end-to-end procedure execution. This approach provides insight into how local layer issues propagate through the system.

8. Experimental Observations

The experimental results confirm that 5G NR system behaviour is highly interdependent across layers. Theoretical procedures described in specifications often exhibit additional constraints when implemented in real systems, making cross-layer understanding critical.

9. Conclusion

This work presents both theoretical and practical analysis of a complete 5G NR end-to-end system implemented using srsRAN and Open5GS. By aligning 3GPP Release-15 concepts with real system behaviour, the project provides a strong foundation for further work in 5G NR protocol development, optimization, and standard-compliant system validation.